

Education

Northeastern University, Boston, MA

Expected Dec 2026

Master of Science in Data Science

Coursework: Database Management, Machine Learning, Natural Language Processing, Unsupervised Learning, MLOps

Experience

Data Scientist Intern

May 2026 – Present

FM, Johnston, RI

- Deployed a **multimodal LLM system** fusing text (inspection notes), **images**, and property documents into structured content classifications, serving **150 reports/day** at **p95 latency below 250ms** via **vLLM** on **GPU-backed Kubernetes** with continuous batching and prefix caching.
- Orchestrated multi-turn model workflows in **Temporal**, layering in automated validation checks to flag **unsupported or non-compliant claims** before human review, reducing manual review burden on downstream engineers.
- Identified **fire code imputation** as a high-signal feature for **GLM**-based loss forecasting across **Fire** and **Boiler & Machinery (B&M)** portfolios, reducing **MAE by 12.5%**.

Data Scientist Co-op

Aug 2025 – Jan 2026

Institute of Experiential AI, Portland, ME

- Trained a **PyTorch deep neural network classifier** on **850K+** insurance policies to flag high-risk and anomalous cases, improving detection accuracy by **20%**.
- Cut batch validation time **75%** (8h → 2h) by architecting an **LLM-assisted validation framework** using Gemini 1.5 Pro on Vertex AI, backed by 120+ automated quality tests.
- Deployed Dockerized Flask APIs on AWS SageMaker, serving over **3,200 predictions/hour** at **p95 latency below 180ms**.
- Migrated and validated **23M+** insurance policy records into PostgreSQL (AWS RDS) via AWS Glue ETL pipelines, achieving **99.97%** source-to-target data integrity.

Machine Learning Intern

Sep 2023 – Aug 2024

Indian Servers, Hyderabad, India

- Fine-tuned DistilBERT using Hugging Face Transformers on 35K labeled support tickets, improving chatbot intent classification accuracy by **20%**.
- Reduced customer response latency **15%** by building LangChain-based prompt engineering workflows with asynchronous inference pipelines using Celery and Redis.

Projects

VIBE: Visual-Intent Bridge via Energy Scoring

Feb 2026

PyTorch, CLIP, Multimodal ML, Streamlit – GitHub – Live Demo

- Built a **multimodal image-text retrieval system** using frozen **CLIP** encoders, learnable projection heads, and **contrastive learning**, achieving **R@1 = 52.8%**, **R@5 = 86.8%**, **R@10 = 93.8%** on MS-COCO.
- Implemented **energy-based scoring** with Langevin MCMC refinement and deployed an interactive **Streamlit** application for real-time multimodal retrieval and visualization.
- Built **ViT attention-rollout** visualizations to interpret which image regions drove each retrieval decision, surfacing failure cases (e.g. black & white, minimalist, or close-up images).

TheraBot: Multimodal NLP Chatbot with Emotion & Sarcasm Detection

2025

PyTorch, Hugging Face Transformers, Flask – GitHub

- Built a **multi-model NLP content-understanding pipeline** combining an attention-based **emotion classifier** (10 classes) and a **sarcasm detector** to score user messages in real time.
- Fine-tuned **DistilGPT2** for context-aware response generation conditioned on detected emotion/sarcasm signals, served via a Flask web application.

Technical Skills

Languages: Python, SQL (PostgreSQL), R, C/C++

Computer Vision & Multimodality: CLIP, Multimodal Retrieval, Energy-Based Models, Vision Transformers, OpenCV

NLP & LLMs: LangChain, LangGraph, RAG, Agentic AI, vLLM, Temporal, Claude, Gemini, Transformers, Content Classification, Prompt Engineering

Machine Learning: Deep Learning, GLMs, Regression, Gradient Boosting, Random Forests, SVMs, Feature Engineering, Model Evaluation

Libraries: PyTorch, Scikit-learn, pandas, NumPy, SciPy, Matplotlib, OpenCV

Data Engineering & Cloud: PostgreSQL, Snowflake, PySpark, Airflow, AWS Glue, AWS (SageMaker, Bedrock, S3, EC2, Lambda, RDS), Google Cloud Platform, Databricks, Docker, Git, GitHub